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*ZC416 Mathematical Foundations for Machine Learning / Data Science

Topic: #4 Eigenvalues & Eigenvectors

MATRIX DECOMPOSITIONS



- We studied vectors and how to manipulate them in preceding lectures.
- Mappings and transformations of vectors can be conveniently described in terms of operations performed by matrices.
- In this lecture we shall study three aspects of matrices: how to summarize matrices, how matrices can be decomposed, and how the decompositions can be used for matrix approximations.

DETERMINANT AND TRACE



A determinant of order $n \times n$ is a scalar associated with a $n \times n$

matrix and is denoted as follows: $\det(\mathbf{A}) = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$

We have a cofactor formula to calculate a determinant of order n : for $n = 1$, we have $\det(\mathbf{A}) = a_{11}$. For $n \geq 2$ we have

$$\begin{aligned} D = \det(\mathbf{A}) &= a_{j1}C_{j1} + a_{j2}C_{j2} + \dots + a_{jn}C_{jn} \\ &= a_{1k}C_{1k} + a_{2k}C_{2k} + \dots + a_{nk}C_{nk} \end{aligned}$$

The first line above represents expansion along the j th row, while the second line represents expansion along the k th column.

COFACTOR FORMULA



- In the preceding slide, the $C_{jk} = (-1)^{j+k} M_{jk}$ where M_{jk} represents the $n - 1$ order determinant of the submatrix of $\mathbf{A}$ obtained by removing the j th row and k th column.
- M_{jk} is called the minor of a_{jk} in D and C_{jk} is called the cofactor of a_{jk} in D .
- Our definition for the determinant in the previous slide shows that the $n \times n$ determinant is defined in terms of $(n - 1) \times (n - 1)$ determinants which in turn are defined in terms of $(n - 2) \times (n - 2)$ determinants recursively.
- Let us examine the computation for a simple 3×3 determinant.

EXAMPLE



Let us compute $D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

For the entries in the second row the minors are $M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$,

$M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$, $M_{23} = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$. The cofactors are $C_{21} = (-1)^{1+2} M_{21} = -M_{21}$, $C_{22} = (-1)^{2+2} M_{22} = M_{22}$, $C_{23} = (-1)^{2+3} M_{23} = -M_{23}$.

Expanding along the second row we can write $D = a_{21} C_{21} + a_{22} C_{22} + a_{23} C_{23}$.

BEHAVIOUR OF DETERMINANT



Theorem: We can state the following for a n th order determinant under elementary row operations: (a) interchanging two rows multiplies the value of the determinant by -1 , and (b) adding a multiple of a row to another row does not change the value of the determinant.

Proof Sketch Let us look at how to prove (a). The proof is by induction. The statement holds for $n = 2$ since $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ whereas $\begin{vmatrix} c & d \\ a & b \end{vmatrix} = bc - ad = -(ad - bc)$. We make the induction hypothesis that the statement is true for all determinants of order $(n-1)$. Let D represent the original determinant and E represent the determinant with rows interchanged.

PROOF SKETCH CONTINUED



Let us expand D and E along a **row that is not interchanged**.
We have

$$D = \sum_{k=1}^{k=n} (-1)^{j+k} a_{jk} M_{jk}$$

$$E = \sum_{k=1}^{k=n} a_{jk} (-1)^{j+k} N_{jk}$$

where N_{jk} in E is obtained by exchanging two rows of the minor M_{jk} in D . M_{jk} and N_{jk} are determinants of order $n - 1$ where one of the determinants has a pair of rows interchanged as compared to the other determinant. Therefore $M_{jk} = -N_{jk}$, and $D = -E$.

PROOF SKETCH CONTINUED



Let us now look at adding multiples of a row to another row.

- Add c times row i to row j . Then we get a new determinant D' whose entries in the j th row has $a_{jk} + ca_{ik}$. Expanding the determinant D' we have

$$D' = \sum_{k=1}^{k=n} (a_{jk} + ca_{ik})(-1)^{j+k} M_{jk} =$$
$$\sum_{k=1}^{k=n} (-1)^{j+k} a_{jk} M_{jk} + \sum_{k=1}^{k=n} (-1)^{j+k} ca_{ik} M_{jk}.$$

- The summation can be written as $D' = D_1 + cD_2$ where $D_1 = D$ and D_2 represents the determinant of a matrix similar to the one we started out with except that rows j and i both have coefficients a_{ik} in them. Thus two rows of D_2 are equal - this makes $D_2 = 0 \rightarrow$ why is this?

PROOF SKETCH CONTINUED



- In part (a) we showed that interchanging two rows will negate the determinant. If we interchange rows i and j we will get the same determinant since the two rows are identical. But one of the determinants must be the negative of the other. This is only possible when both determinants are zero. Thus $D_2 = 0$ and $D' = D$.
- **Bottom line** → Adding a multiple of one row to another row does not change the determinant.
- This will lead us to our next result.

ANOTHER RESULT



Theorem An $n \times n$ matrix $\mathbf{A}$ has rank n if and only if its determinant is not equal to zero.

Proof sketch: $\mathbf{A}$ has full rank $\implies \det(\mathbf{A}) \neq 0$: Gaussian elimination reduces $\mathbf{A}$ to upper triangular matrix $\mathbf{U} = (U_{ij})$ whose determinant is the product of all the elements U_{ii} . But we know that $\det(\mathbf{A}) = (-1)^s \det(\mathbf{U})$ where s is the number of row interchanges performed during Gaussian elimination. Since $\mathbf{A}$ has full rank, the columns of $\mathbf{A}$ are linearly independent and the only solution to $\mathbf{A}\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$. The system $\mathbf{A}\mathbf{x} = \mathbf{0}$ has the same set of solutions as $\mathbf{U}\mathbf{x} = \mathbf{0}$, so this means $\mathbf{U}$ has only pivot columns. The pivots are all the elements U_{ii} . The product of all pivots is non-zero, and hence $\det(\mathbf{U}) = \det(\mathbf{A}) \neq 0$.

ANOTHER RESULT

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$\det(\mathbf{A}) \neq 0 \implies \mathbf{A}$ has full rank: If the determinant of $\mathbf{A}$ is non-zero, $\det(\mathbf{A}) = (-1)^s \det(\mathbf{U}) = \prod_{i=1}^n U_{ii}$ means that all the U_{ii} are non-zero. Therefore all the columns of $\mathbf{U}$ are pivot columns with the pivots being the U_{ii} . The pivot columns are all linearly independent, so the only solution to $\mathbf{U}\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$. This is also the only solution to $\mathbf{A}\mathbf{x} = \mathbf{0}$ which means $\mathbf{A}$ has full rank.

- The trace of a $n \times n$ square matrix $\mathbf{A}$ is defined as $tr(\mathbf{A}) = \sum_{i=1}^n a_{ii}$, i.e the sum of the diagonal elements of $\mathbf{A}$.
- The trace has the following properties:
 - ▶ $tr(\mathbf{A} + \mathbf{B}) = tr(\mathbf{A}) + tr(\mathbf{B})$, for $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$
 - ▶ $tr(\alpha \mathbf{A}) = \alpha tr(\mathbf{A})$, $\alpha \in \mathbb{R}$
 - ▶ $tr(\mathbf{I}_n) = n$
 - ▶ $tr(\mathbf{AB}) = tr(\mathbf{BA})$ for $\mathbf{A} \in \mathbb{R}^{n \times k}$, $\mathbf{B} \in \mathbb{R}^{k \times n}$.
- The proofs of these properties are not difficult.

CHARACTERISTIC POLYNOMIAL



- For $\lambda \in \mathbb{R}$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$ we can define $p_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I})$ and show that it can be written as $c_0 + c_1\lambda + \dots + c_{n-1}\lambda^{n-1} + (-1)^n\lambda^n$ where $c_0, c_1, \dots, c_{n-1} \in \mathbb{R}$.
- We can show that $c_0 = \det(\mathbf{A})$ and $c_{n-1} = (-1)^{n-1} \text{tr}(\mathbf{A})$
- To see that $c_0 = \det(\mathbf{A})$, set $\lambda = 0$ in $\det(\mathbf{A} - \lambda \mathbf{I})$ to get $p_{\mathbf{A}}(0) = \det(\mathbf{A}) = c_0$
- The formula for c_{n-1} takes a little bit of work - let us expand a 3×3 determinant

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix}$$

CHARACTERISTIC POLYNOMIAL



- Expanding the determinant along the first row we see that

the $(a_{11} - \lambda)C_{11}$ term contains the product $\prod_{i=1}^{i=3} (a_{ii} - \lambda)$

which contains the powers λ^3 and λ^2 . The other contributors to the determinant i.e $a_{12}C_{12}$ and $a_{13}C_{13}$ expand into terms where the maximum power of $\lambda = 1$.

- Carrying this analogy over to the general case of $n > 3$ we see that expanding along the first row the first contributor to

the determinant will have the term $\prod_{i=1}^{i=n} (a_{ii} - \lambda)$ and

subsequent contributors will have a maximum power of λ^{n-2} as the minors for each such contributor will kill off a term containing λ in a given row and column.

CHARACTERISTIC POLYNOMIAL



- Thus in the determinant expansion to obtain the characteristic polynomial we see that coefficient to λ^{n-1}

can only come from the expansion of $\prod_{i=1}^{i=n} (a_{ii} - \lambda)$ and can

be seen to be $(-1)^{n-1} \sum_{i=1}^{i=n} a_{ii} = (-1)^{n-1} \text{tr}(\mathbf{A})$.

- As a corollary to this argument we can see that the coefficient to λ^n in the characteristic polynomial is $(-1)^n$
- We will use the characteristic polynomial to compute eigenvalues and eigenvectors.

EIGENVALUES AND EIGENVECTORS



- Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix. Then $\lambda \in \mathbb{R}$ is an eigenvalue of $\mathbf{A}$ and $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$ is the corresponding eigenvector of λ if $\mathbf{Ax} = \lambda\mathbf{x}$. This equation is called the eigenvalue equation.
- The following statements are equivalent:
 - ▶ λ is an eigenvalue of $\mathbf{A} \in \mathbb{R}^{n \times n}$.
 - ▶ There exists an $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$ with $\mathbf{Ax} = \lambda\mathbf{x}$, or equivalently, $(\mathbf{A} - \lambda\mathbf{I}_n)\mathbf{x} = \mathbf{0}$ can be solved non-trivially, i.e., $\mathbf{x} \neq \mathbf{0}$.
 - ▶ $\text{rank}(\mathbf{A} - \lambda\mathbf{I}_n) < n$.
 - ▶ $\det(\mathbf{A} - \lambda\mathbf{I}_n) = 0$.
- If $\mathbf{x}$ is an eigenvector corresponding to a particular eigenvalue λ , $c\mathbf{x}$, $c \in \mathbb{R} \setminus \mathbf{0}$ is also an eigenvector.

EIGENVALUES AND EIGENVECTORS - EXAMPLE

- Consider the matrix $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. The characteristic polynomial $\det(\mathbf{A} - \lambda \mathbf{I}) = (1 - \lambda)^2 - 1$ and setting it to zero gives us the roots of the characteristic polynomial: $(1 - \lambda)^2 - 1 = 0$ has roots $\lambda = 2, 0$.
- What are the eigenvectors? For $\lambda = 0$ we solve for $\mathbf{A}\mathbf{x} = 0\mathbf{x}$, so we find the nullspace of the matrix $\mathbf{A}$. Using Gaussian elimination we convert $\mathbf{A}\mathbf{x} = \mathbf{0}$ to $\mathbf{U}\mathbf{x} = \mathbf{0}$ where $\mathbf{U} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. Thus we discover the eigenvector $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ for $\lambda = 0$.
- Similarly we discover the eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ for $\lambda = 2$.

EIGENVALUES AND EIGENVECTORS - EXAMPLE

- The general procedure to find eigenvalues and eigenvectors is to first find the roots of the characteristic polynomials and then find the nullspaces of the matrices $\mathbf{A} - \lambda \mathbf{I}$ for the different roots λ .
- Does every $n \times n$ matrix have a full set of eigenvectors, i.e. n eigenvectors?
- Look at $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. What are its eigenvalues and eigenvectors?
- **Point to ponder** Looking at the equation $\mathbf{Ax} = \lambda \mathbf{x}$ it seems that the action of $\mathbf{A}$ on $\mathbf{x}$ is to preserve the direction of $\mathbf{x}$ but scale it up or down according to λ . Does this mean that a rotation matrix has no eigenvalues and eigenvectors?

SOME ADDITIONAL PROPERTIES



- λ is an eigenvalue of $\mathbf{A}$ if and only if λ is a root of the characteristic polynomial $p_{\mathbf{A}}(\lambda)$ of $\mathbf{A}$. This can be easily seen as a consequence of the definition of $p_{\mathbf{A}}(\lambda)$.
- For $\mathbf{A} \in \mathbb{R}^{n \times n}$, the set of eigenvectors corresponding to an eigenvalue λ spans a subspace of $\mathbb{R}^n$ called the Eigenspace of $\mathbf{A}$ with respect to λ and is denoted by E_{λ} .
- The set of all eigenvalues of $\mathbf{A}$ is called the spectrum of $\mathbf{A}$.
- Look at the eigenvalues and eigenspace of the $n \times n$ identity matrix $\mathbf{I}_n$. It has one eigenvalue $\lambda = 1$ and the eigenspace is $\mathbb{R}^n$. Every canonical vector is a basis vector for the eigenspace.

SOME ADDITIONAL PROPERTIES



- A matrix and its transpose have the same eigenvalues. To see this, first note that $\det(\mathbf{A}) = \det(\mathbf{A}^T)$. Then $\det(\mathbf{A} - \lambda \mathbf{I}) = \det((\mathbf{A} - \lambda \mathbf{I})^T) = \det(\mathbf{A}^T - \lambda \mathbf{I}^T) = \det(\mathbf{A}^T - \lambda \mathbf{I})$. The last expression in the chain of equalities is the characteristic polynomial for $p_{\mathbf{A}^T}(\lambda)$. Thus we have $p_{\mathbf{A}}(\lambda) = p_{\mathbf{A}^T}(\lambda)$ which means the characteristic polynomials are equal and so the roots of the polynomials or the eigenvalues must be equal.
- The eigenspace E_λ is the nullspace of $\mathbf{A} - \lambda \mathbf{I}$.
- Symmetric, positive-definite matrices always have positive, real eigenvalues.

SOME THEOREMS



- The eigenvectors $\mathbf{x}_1, \mathbf{x}_2 \dots \mathbf{x}_n$ of a $n \times n$ matrix $\mathbf{A}$ with n **distinct** eigenvalues are linearly independent $\rightarrow$ why?
- Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ we can show that $\mathbf{A}^T \mathbf{A} \in \mathbb{R}^{n \times n}$ is a symmetric, positive-definite matrix when the rank of $\mathbf{A} = n$. Why is this true? Clearly $\mathbf{A}^T \mathbf{A}$ is a symmetric matrix and it is positive definite since $\mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} = \|\mathbf{A} \mathbf{x}\|^2 > 0 \quad \forall \mathbf{x} \in \mathbb{R}^n \setminus 0$ since the nullspaces of $\mathbf{A}^T \mathbf{A}$ and $\mathbf{A}$ are the same, and $\mathbf{A}$ is a full column rank matrix.
- The matrix $\mathbf{A}^T \mathbf{A}$ is important in machine learning since it figures in the least-squares solution to a data matrix represented as $\mathbf{A}$ where n represents the number of features and m is the number of data vectors.

SPECTRAL THEOREM



Theorem: If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric there exists an orthonormal basis of the corresponding vector space V consisting of the eigenvectors of $\mathbf{A}$, and each eigenvalue is real.

Proof: We will not attempt a full proof of this theorem but provide some intuitions about why it is true. The theorem relies on the following three statements, shown in the next slide.

SPECTRAL THEOREM



- All roots of the characteristic polynomial $p_A(\lambda)$ are real.
- For each eigenvalue λ we can compute an orthonormal basis for its eigenspace. We can string together the orthonormal bases for the different eigenvalues of $\mathbf{A}$ to come up with the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots$
- The dimension of the eigenspace E_λ , called its geometric multiplicity, is the same as the algebraic multiplicity of λ which is the number of times λ appears as a root of the characteristic polynomial.
- All the basis vectors from the different Eigenspaces combine to provide an orthonormal basis for $\mathbb{R}^n$.

COMPLEX VECTORS



- In the old formulation with real vectors, length-squared according to the Euclidean norm was $x_1^2 + x_2^2 + \dots x_n^2$. If the x_i are complex we should take length-squared to be $|x_1|^2 + |x_2|^2 + \dots |x_n|^2$ where $|\cdot|$ denotes modulus. For the complex number $a + bi$, the modulus is $\sqrt{(a + bi)(a - bi)} = \sqrt{a^2 + b^2}$.
- For complex vectors we would like to preserve the idea as possible that $\|x\|^2 = \mathbf{x}^T \mathbf{x}$. If we keep the old definition of inner product for complex vectors we will not get a real number as length as shown in the next bullet.
- Let $\mathbf{x} = \begin{bmatrix} 1 + i \\ 2 + i \end{bmatrix}$. We have $\mathbf{x}^T \mathbf{x} = (1 + i)^2 + (2 + i)^2 = 1 + 2i + i^2 + 4 + 4i + i^2 = 6i + 3$.

HERMITIAN MATRICES



- We modify the inner product between two complex vectors $\mathbf{x}$ and $\mathbf{y}$ to $\mathbf{x}^H \mathbf{y}$, where $\mathbf{x}^H = \overline{\mathbf{x}}^T$.
- Now $\mathbf{x}^H \mathbf{x} = \overline{x}_1 x_1 + \dots \overline{x}_n x_n = \|\mathbf{x}\|^2$ according to the new definition of length.
- A Hermitian matrix is a generalization of a symmetric matrix.
- Instead of requiring $\mathbf{A}^T = \mathbf{A}$, we say a matrix is Hermitian if it is equal to its conjugate-transpose, ie $\mathbf{A}$ is a Hermitian matrix if $\mathbf{A}^H = \mathbf{A}$ or $\overline{\mathbf{A}}^T = \mathbf{A}$
- As an example consider the matrix $\mathbf{A} = \begin{bmatrix} 1 & 3-i \\ 3+i & 4 \end{bmatrix}$. It is a Hermitian matrix since $\mathbf{A}^H = \begin{bmatrix} 1 & 3-i \\ 3+i & 4 \end{bmatrix} = \mathbf{A}$.

SPECTRAL THEOREM

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We shall now show that all eigenvalues for a symmetric matrix are real. Let $\mathbf{Ax} = \lambda\mathbf{x}$. Then premultiplying with $\mathbf{x}^H$ on both sides we have $\mathbf{x}^H \mathbf{Ax} = \lambda \mathbf{x}^H \mathbf{x}$.

Now $\mathbf{x}^H \mathbf{Ax}$ is a 1×1 matrix. Taking the Hermitian of this matrix we have $(\mathbf{x}^H \mathbf{Ax})^H = \mathbf{x}^H \mathbf{A}^H \mathbf{x} = \mathbf{x}^H \mathbf{Ax}$, so the Hermitian of the matrix is itself which means that the matrix is real.

On the right hand side we note that $\mathbf{x}^H \mathbf{x}$ is real, so this means that λ must be real.

SPECTRAL THEOREM



Let us show that eigenvectors belonging to different eigenvalues are orthogonal. Let $\mathbf{Ax} = \lambda\mathbf{x}$ and $\mathbf{Ay} = \mu\mathbf{y}$. Then we have

$$\begin{aligned} \mathbf{y}^H \mathbf{Ax} &= \lambda \mathbf{y}^H \mathbf{x} \\ \mathbf{x}^H \mathbf{Ay} &= \mu \mathbf{x}^H \mathbf{y} \end{aligned}$$

But $\mathbf{x}^H \mathbf{Ay} = (\mathbf{y}^H \mathbf{A}^H \mathbf{x})^H = (\mathbf{y}^H \mathbf{Ax})^H = \lambda \mathbf{x}^H \mathbf{y}$. We already know that $\mathbf{x}^H \mathbf{Ay} = \mu \mathbf{x}^H \mathbf{y}$. This means $\lambda \mathbf{x}^H \mathbf{y} = \mu \mathbf{x}^H \mathbf{y}$. Since $\lambda \neq \mu$, this must mean $\mathbf{x}^H \mathbf{y} = 0$.

This shows that eigenvectors corresponding to different eigenvalues are orthogonal.

SPECTRAL THEOREM



- So we see that the eigenvalues of a symmetric matrix are real and eigenvectors belonging to different eigenvalues are orthogonal.
- This suggests that one can string together all the orthonormal bases for the different eigenvalues and get an orthonormal basis for $\mathbb{R}^n$.
- But who is to say that when we string together the basis vectors for all the eigenvalues, we will have enough vectors to describe $\mathbb{R}^n$? We need n basis vectors and might end up having fewer than n vectors.
- If the eigenvalues are all different, we can see that we will indeed have enough basis vectors. But what about when there are repeating eigenvalues?

SPECTRAL THEOREM



- We need one more piece to complete the puzzle and show that we will have enough eigenvectors to complete the orthonormal basis - this part we shall not prove!
- As a consequence of the spectral theorem we can write a real symmetric matrix $\mathbf{A}$ as $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ where $\mathbf{Q}$ is an orthonormal basis (think orthonormal basis vectors for example), and $\mathbf{\Lambda}$ is a diagonal matrix consisting of non-zero entries only along the diagonal.
- The spectral theorem can be used in a machine learning context since we can take the data matrix $\mathbf{A}$ and create a symmetric matrix out of it - $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}\mathbf{A}^T$ which are both used in Singular-Value Decomposition and PCA.

TRACE AND EIGENVALUES



- We can show that the sum of the eigenvalues of a matrix is equal to the trace of the matrix, i.e. $\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii}$. To see why this is true, note that the characteristic polynomial $p_A(\lambda)$ can be written as $\prod_{i=1}^n (\lambda_i - \lambda)$. The coefficient to λ^{n-1} in this expansion is $(-1)^{n-1} \sum_{i=1}^n \lambda_i$. Early on in this lecture we showed from a direct expansion of the determinant that the coefficient of λ^{n-1} is $(-1)^{n-1} \sum_{i=1}^n a_{ii}$. Thus we have our result.
- The product of all eigenvalues is the determinant of the matrix, i.e. $\det(\mathbf{A}) = \prod_{i=1}^n \lambda_i$. To see why this is true, let us once again look at the factorisation of $p_A(\lambda)$ as $\det(\mathbf{A} - \lambda \mathbf{I}) = p_A(\lambda) = \prod_{i=1}^n (\lambda_i - \lambda)$. Setting $\lambda = 0$ in this equation gives the result.

CHOLESKY DECOMPOSITION



Theorem A symmetric, positive-definite matrix A can be factorized into a product $A = LL^T$ where L is a lower-triangular matrix with positive elements.

For an example 3×3 matrix we can write

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

We can solve for the elements of the lower triangular matrix to get $l_{11} = \sqrt{a_{11}}$, $l_{22} = \sqrt{a_{22} - l_{21}^2}$, $l_{33} = \sqrt{a_{33} - (l_{31}^2 + l_{32}^2)}$.

For the elements below the diagonal we have $l_{21} = \frac{a_{21}}{l_{11}}$, $l_{31} = \frac{a_{31}}{l_{11}}$ and $l_{32} = \frac{a_{32} - l_{31}l_{21}}{l_{22}}$.

AN APPLICATION OF CHOLESKY DECOMPOSITION

- In the Data Science / Machine Learning context, distributions on data are frequently multivariate Gaussian.
- Multivariate Gaussian distributions are governed by a covariance matrix which is symmetric, positive-definite.
- We may need to draw samples from such distributions which is where the Cholesky decomposition finds an important application.
- To generate a sample from a multivariate Gaussian distribution, we factor the covariance matrix into its Cholesky factor $\mathbf{L}$, generate a Gaussian random vector $\mathbf{x}$ on independent variables which is easy to do, and compute $\mathbf{Lx}$ which will be a sample according to the required distribution.



THANK YOU